Robert Boyce

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Academic Degrees

Imperial College London 09/2023 - present *(expected 08/2027)*
PhD in Mathematics of Random Systems (in progress).
Supervised by Dr Eyal Neuman.

University of Warwick 09/2022 - 09/2023
MSc in Mathematical Finance (with Distinction).
Dissertation supervised by Professor Martin Herdegen.
Awarded university scholarship worth 50% of course fee.

London School of Economics and Political Science 09/2019 - 07/2022
BSc in Financial Mathematics and Statistics (with First-Class honours).

Publications and preprints

[1] Barzykin A., Boyce R., Neuman E., Tuschmann S. **Optimal Execution with Passive Market Impact.** arXiv: 2607.28323. *(submitted to Mathematical Finance).* 07/2026

[2] Boyce R., Neuman E. **Competition in Dealer Markets with Internalisation and Externalisation.** arXiv: 2606.06413. *(submitted to SIAM Journal on Financial Mathematics).* 06/2026

[3] Barzykin A., Boyce R., Neuman E. **FX Market Making with Internal Liquidity**. Risk Cutting Edge. 04/2026

[4] Boyce R., Herdegen M., Sánchez-Betancourt L. **Market Making with Exogenous Competition**. SIAM Journal on Financial Mathematics. 06/2025

[5] Barzykin A., Boyce R., Neuman E. **Unwinding Toxic Flow with Partial Information**. arXiv: 2407.04510 *(submitted to Quantitative Finance).* 07/2024

Industrial experience

Farallon Capital Management, *Equity Volatility Intern* 02/2026 – 07/2026
Designed and implemented in-house pricing workflows for exotic options and convertible bonds.
Used time series models for forecasting portfolio risk across the firm.
Researched bias arising from backtesting methods.

HSBC FX Market Making, *Quantitative Finance Summer Associate* 07/2025 - 10/2025
Wrote *FX Market Making with Internal Liquidity*, calibrated using proprietary trade data. Published as a Risk Cutting Edge article. Developed and deployed a forward-looking analytics tool for the FX desk.

Teaching experience

Numerical Methods for Finance 01/2026 – 02/2026
Teaching assistant running seminars for numerical methods for finance. The course covers tree methods, PDE methods, and Fourier methods for option pricing.

Referee/reviewer

Mathematical Finance, Quantitative Finance

Talks

- UCSB Financial Mathematics Seminar 09/2026
- UCLA Mathematical and Statistical Finance Seminar 09/2026
- Columbia University Mathematical Finance Seminar 09/2026
- 13th Bachelier World Congress 2026, University of Bologna 07/2026
- ETH – Hong Kong – Imperial workshop on mathematical finance, ETH Zurich 06/2026
- XXVII Workshop on Quantitative Finance, University of Bergamo 03/2026
- Mathematics of Random Systems CDT workshop, Imperial College London 11/2025
- SIAM Conference on Financial Mathematics and Engineering (FM25), Miami 07/2025
- ETH – Hong Kong – Imperial Mathematical Finance Workshop, Hong Kong Polytechnic University 04/2025
- Dolomites Winter School on Optimal Transport 01/2025
- London-Oxford-Warwick Mathematical Finance Workshop, University of Oxford 01/2025
- Algo-trading & DEFI Methods and Technologies Workshop, Politecnico di Milano 11/2024
- 7th Berlin Workshop on Mathematical Finance for Young Researchers 09/2024
- 12th Bachelier World Congress 2024, Fundação Getulio Vargas, Rio de Janeiro 07/2024
- ETH – Hong Kong – Imperial Mathematical Finance Workshop, Imperial College London 06/2024
- Mathematics of Random Systems CDT spring retreat, Missenden Abbey 04/2024

Awards

- 13th Bachelier World Congress 2026 travel and accommodation support (2026)
- ETH – Hong Kong – Imperial Mathematical Finance Workshop, Zurich, accommodation support (2026)
- G-Research Imperial College travel grant (2025)
- ETH – Hong Kong – Imperial Mathematical Finance Workshop, Hong Kong accommodation support (2025)
- Dolomites Winter School accommodation support (2025)
- G-Research Imperial College travel grant (2024)
- Studentship in the EPSRC CDT in Mathematics of Random Systems (2023-2027)
- MSc Mathematical Finance scholarship worth 50% of course fee (2022-2023)